Robust Indoor Mobile Robot Localization Using Maximum Correntropy Criterion UKF

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***Abstract*—Accurate indoor localization for autonomous mobile robots (AMRs) is challenging due to complex environments that often introduce non-Gaussian noise and severe outliers in Ultra-Wideband (UWB) distance measurements, degrading the performance of traditional filtering techniques. To address this issue, we propose a Sensor Fusion-based Maximum Correntropy Criterion Unscented Kalman Filter (SF-MCC-UKF). The algorithm fuses synchronized UWB, wheel-encoder, and inertial data for continuous position estimation, while integrating a correntropy-based mechanism into the standard UKF to dynamically mitigate non-Gaussian noise and outliers. To validate its performance, the SF-MCC-UKF is compared with four baselines: Least Squares Estimation (LSE), Sensor Fusion-based Kalman Filter (SF-KF), Maximum Correntropy Criterion Unscented Kalman Filter (MCC-UKF), and Sensor Fusion-based Unscented Kalman Filter (SF-UKF). The evaluation is conducted under identical conditions, including the same anchor layout, corrected Single-Sided Two-Way Ranging (SS-TWR) measurements, trajectory, and time base. Experimental results demonstrate superior accuracy and robustness, with the SF-MCC-UKF achieving a positioning Root Mean Square Error (RMSE) of 146.9 mm, improving by 33.1%, 23.5%, 21.2%, and 20.2% compared to the LSE, SF-KF, MCC-UKF, and SF-UKF baselines, respectively.**

***Index Terms*—Ultra-wideband, indoor localization, autonomous mobile robot, least squares estimation, Kalman filter, unscented Kalman filter, sensor fusion, maximum correntropy criterion.**

I. Introduction

AMRs are increasingly deployed in factories, warehouses, and service environments for material transportation and automated inspection [1]. Reliable operation requires continuous indoor localization, where Global Navigation Satellite System signals are generally unavailable. Alternative wireless technologies such as Wi-Fi [2] and Bluetooth [6] often provide insufficient accuracy for compact industrial workspaces.

UWB offers low-latency, centimeter-level indoor positioning, with ideal-LOS errors modeled as zero-mean Gaussian noise [4], [5]. However, practical indoor workspaces, even with clear LOS, contain floors, ceilings, and objects causing near-field multipath reflections. These can distort signals and trick the receiver into locking onto reflected paths rather than the direct one, shifting the error distribution from pure Gaussian to heavy-tailed non-Gaussian noise with severe outliers [6].

Previous studies provide the foundation for the methods investigated in this work. Zhou [7] presented an efficient least-squares formulation for UWB-based mobile robot localization. Xiong *et al.* [8] introduced the MCC to improve Time-of-Arrival (TOA) localization in the presence of abnormal measurement errors, while Wang *et al.* [9] extended correntropy-based estimation to mitigate bias in Time-Difference-of-Arrival (TDOA) measurements. MCC has also been integrated with UKF to improve nonlinear state estimation and enhance robustness against heavy-tailed measurement noise in UWB localization systems [10], [11], [12]. However, these studies generally investigate their methods under different datasets or experimental settings, making direct performance comparisons difficult. In addition, relatively few studies have combined synchronized UWB distance measurements with wheel-encoder and inertial data in an MCC-based UKF system for indoor AMR localization.

To address these challenges, we combine correntropy-based UWB weighting with vehicle kinematics in a unified sensor-fusion framework.

The main contributions of this paper are summarized as follows:

1) An SF-MCC-UKF framework is developed for AMR localization by integrating synchronized UWB ranges, wheel-encoder speed, and IMU yaw-rate measurements. The framework combines nonlinear sigma-point estimation with componentwise correntropy weighting during the UWB measurement update, reducing the influence of large ranging residuals on the position estimate.
2) A unified evaluation framework is established to compare UWB-only and sensor-fusion localization methods under identical conditions. LSE, SF-KF, MCC-UKF, SF-UKF, and the proposed SF-MCC-UKF are evaluated using the same anchor configuration, corrected SS-TWR measurements, reference trajectory, initialization procedure, and time base. The simulation examines robustness under synthetically injected non-Gaussian UWB outliers, while the physical experiment is conducted in the indoor LOS environment.

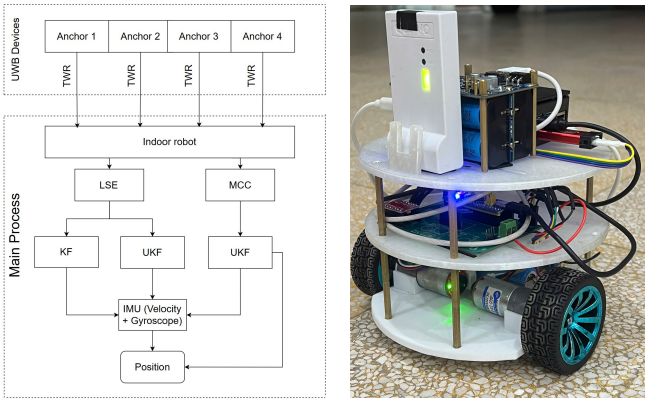
The remainder of this paper is organized as follows. Section II describes the localization system and hardware configuration. Section III presents LSE, MCC, and SF-MCC-

UKF. Section IV reports the experimental setup and results. Section V concludes the paper.

II. LOCALIZATION SYSTEM

The positioning system is built using compact DWM1001 modules integrating a UWB transceiver, an ARM Cortex-M4 processor, and Bluetooth Low Energy (BLE). Four modules are configured as fixed reference nodes (Anchors) at known locations, while one operates as a mobile target (Tag) equipped on the AMR. The anchors and tag perform continuous Two-Way Ranging (TWR) by exchanging timestamped packets to estimate distances from time-of-flight information. All anchors interact with the tag simultaneously, supporting a measurement update rate of up to 160 Hz, which enables robust and fast measurements for high-accuracy real-time positioning. As illustrated in Fig. 1(a), the system's data flow is conceptually divided into a *UWB Devices* layer and a *Main Process* layer. The raw TWR distances from the anchors are provided into the indoor robot and routed into two distinct algorithmic tracks: LSE and MCC. Within the Main Process, the distance data from LSE track is filtered through KF or UKF, before being fused with kinematic measurements (velocity and gyroscope) from the robot's IMU. Meanwhile, the data from MCC track branches into two paths: it is either processed directly by UKF to output the position, generating the MCC-UKF baseline, or fused with IMU's kinematics and passed through UKF to form the proposed SF-MCC-UKF algorithm. As a result, these parallel pipelines converge to output the final estimated position.

The AMR itself, as shown in Fig. 1(b) is structurally divided into two primary processing blocks. The sensor processing block, comprising an STM32F103C8T6 microcontroller, a BNO055 9-axis sensor, and a magnetic encoder, acquires internal data and controls the vehicle. Meanwhile, the localization processing block, which consists of a Raspberry Pi 4 and the DWM1001 module, reads the UWB distance data, fuses it with the sensor block's information to estimate the current location, and subsequently calculates the return trajectory.



(a)

(b)

Fig. 1. System architecture: (a) UWB system model; (b) Our AMR.

III. LOCALIZATION METHODS

A. Least Squares Estimation Initialization

Because the anchors are installed at a height h m above the ground, the measured tag-to-anchor distance $d_{i,tag}$ must be converted to the two-dimensional ground-plane distance d_i by the Pythagorean relation

$$d_i = \sqrt{d_{i,tag}^2 - h^2}. \tag{1}$$

The Least Squares Estimation (LSE) algorithm is then used to estimate the initial two-dimensional position from the four corrected distances d_i. This initial position is used as a reliable starting state for the recursive estimators, helping the filters converge faster. The LSE formulation [7] is

$$\mathbf{A}\mathbf{x}_{LS} = \mathbf{b}, \tag{2}$$

where

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \\ \vdots & \vdots \\ x_N - x_1 & y_N - y_1 \end{bmatrix}, \quad \mathbf{x}_{LS} = \begin{bmatrix} x_{LS} \\ y_{LS} \end{bmatrix}, \\ \mathbf{b} &= \frac{1}{2} \begin{bmatrix} x_2^2 + y_2^2 - d_2^2 - (x_1^2 + y_1^2 - d_1^2) \\ x_3^2 + y_3^2 - d_3^2 - (x_1^2 + y_1^2 - d_1^2) \\ \vdots \\ x_N^2 + y_N^2 - d_N^2 - (x_1^2 + y_1^2 - d_1^2) \end{bmatrix}. \end{aligned} \tag{3}$$

The estimated coordinate is obtained by solving

$$\mathbf{x}_{LS} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}. \tag{4}$$

B. Maximum Correntropy Criterion

Correntropy is a localized similarity measure between two random variables and is particularly useful when the error distribution is non-Gaussian or contains impulsive outliers. Following the formulation in [10], the correntropy between random variables X and Y is defined as

$$V(X, Y) = E[\kappa(X, Y)], \tag{5}$$

where $E[\cdot]$ denotes expectation and $\kappa(\cdot, \cdot)$ is a positive-definite kernel. Using the Gaussian kernel,

$$G_\sigma(e) = \exp\left(-\frac{e^2}{2\sigma^2}\right), \tag{6}$$

where $e = x - y$ and σ is the kernel bandwidth. For N samples, correntropy is estimated by

$$\widehat{V}(X, Y) = \frac{1}{N} \sum_{i=1}^{N} G_\sigma(e_i). \tag{7}$$

The Maximum Correntropy Criterion (MCC) obtains the estimate that maximizes the sample correntropy,

$$\widehat{\boldsymbol{\xi}} = \arg\max_{\boldsymbol{\xi}} \frac{1}{N} \sum_{i=1}^{N} G_\sigma(e_i(\boldsymbol{\xi})). \tag{8}$$

Because the Gaussian kernel rapidly approaches zero for large residuals, distance measurements affected by severe errors contribute much less to the estimate than ones with small residuals. The kernel bandwidth controls the trade-off between robustness and sensitivity: a smaller σ rejects outliers more strongly, whereas a larger σ makes the criterion approach a quadratic-error estimator [10].

C. Sensor Fusion Maximum Correntropy Criterion Unscented Kalman Filter

SF-MCC-UKF combines the same direct encoder/IMU speed–yaw-rate update used by SF-UKF with nonlinear sigma-point estimation and correntropy weighting for the UWB distances update. For a residual e, the Gaussian correntropy kernel is

$$G_\sigma(e) = \exp\left(-\frac{e^2}{2\sigma^2}\right), \tag{9}$$

where σ is the kernel bandwidth. Large residuals receive weights close to zero, whereas small residuals retain weights close to one. Hence, maximizing correntropy is less sensitive to impulsive or heavy-tailed errors than minimizing a purely quadratic loss [10], [12].

The SF-MCC-UKF state is

$$\mathbf{x}_k = \begin{bmatrix} x_k & y_k & v_k & \omega_k \end{bmatrix}^T, \tag{10}$$

where v_k is the scalar forward speed and ω_k is the yaw rate. An auxiliary heading θ_k is maintained outside the Kalman state and is used by the motion model.

Let $L = 4$ be the state dimension and

$$\lambda_u = \alpha_u^2(L + \kappa_u) - L. \tag{11}$$

The unscented weights are

$$\begin{aligned} W_m^{(0)} &= \frac{\lambda_u}{L+\lambda_u}, \\ W_c^{(0)} &= \frac{\lambda_u}{L+\lambda_u} + 1 - \alpha_u^2 + \beta_u, \\ W_m^{(i)} &= W_c^{(i)} = \frac{1}{2(L+\lambda_u)}, \quad i = 1, \ldots, 2L. \end{aligned} \tag{12}$$

For prediction, sigma points are generated from $\widehat{\mathbf{x}}_{k-1}$ and $\mathbf{P}_{k-1}$ and propagated through

$$f(\boldsymbol{\chi}, \theta_{k-1}) = \begin{bmatrix} x + v\cos(\theta_{k-1} + \omega\Delta t)\Delta t \\ y + v\sin(\theta_{k-1} + \omega\Delta t)\Delta t \\ v \\ \omega \end{bmatrix}. \tag{13}$$

The predicted mean, process-propagated covariance, and process-noise-scaled covariance are

$$\widehat{\mathbf{x}}_k^- = \sum_{i=0}^{2L} W_m^{(i)} \boldsymbol{\chi}_{k|k-1}^{(i)}, \tag{14}$$

$$\mathbf{P}_k^p = \sum_{i=0}^{2L} W_c^{(i)} (\boldsymbol{\chi}_{k|k-1}^{(i)} - \widehat{\mathbf{x}}_k^-)(\boldsymbol{\chi}_{k|k-1}^{(i)} - \widehat{\mathbf{x}}_k^-)^T, \tag{15}$$

$$\mathbf{P}_k^- = \mathbf{P}_k^p + \lambda_{k-1}\mathbf{Q},$$

where λ_{k-1} adaptively scales the process-noise covariance $\mathbf{Q}$. The auxiliary heading is then propagated as

$$\theta_k = \theta_{k-1} + \widehat{\omega}_k^- \Delta t. \tag{16}$$

The encoder and IMU provide a direct speed/yaw-rate observation,

$$\mathbf{z}_{s,k} = \begin{bmatrix} v_{enc,k} \\ \omega_{z,k} \end{bmatrix}, \qquad \mathbf{H}_s = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{17}$$

With sensor covariance $\mathbf{R}_s$, the Gaussian speed/yaw-rate residual, covariance, gain, state, and covariance are

$$\mathbf{r}_{s,k} = \mathbf{z}_{s,k} - \mathbf{H}_s\widehat{\mathbf{x}}_k^-, \tag{18}$$

$$\mathbf{S}_{s,k} = \mathbf{H}_s\mathbf{P}_k^-\mathbf{H}_s^T + \mathbf{R}_s, \qquad \mathbf{K}_{s,k} = \mathbf{P}_k^-\mathbf{H}_s^T\mathbf{S}_{s,k}^{-1}, \tag{19}$$

$$\widehat{\mathbf{x}}_k^* = \widehat{\mathbf{x}}_k^- + \mathbf{K}_{s,k}\mathbf{r}_{s,k}, \tag{20}$$

$$\mathbf{P}_k^* = (\mathbf{I} - \mathbf{K}_{s,k}\mathbf{H}_s)\mathbf{P}_k^-. \tag{21}$$

Only the subsequent UWB distances update uses MCC weighting.

The UWB update generates $2L+1$ sigma points from the intermediate sensor-fusion state,

$$\boldsymbol{\chi}_k^{(0)} = \widehat{\mathbf{x}}_k^*, \tag{22}$$

$$\boldsymbol{\chi}_k^{(i)} = \widehat{\mathbf{x}}_k^* + \left[\sqrt{(L+\lambda_u)\mathbf{P}_k^*}\right]_i, \tag{23}$$

$$\boldsymbol{\chi}_k^{(i+L)} = \widehat{\mathbf{x}}_k^* - \left[\sqrt{(L+\lambda_u)\mathbf{P}_k^*}\right]_i, \tag{24}$$

for $i = 1, \ldots, L$.

For anchor j at (x_{a_j}, y_{a_j}), the ground-plane range function is

$$h_j(\mathbf{x}) = \sqrt{(x - x_{a_j})^2 + (y - y_{a_j})^2}. \tag{25}$$

Each sigma point is transformed into the four-dimensional range space,

$$\mathbf{z}_k^{(i)} = \begin{bmatrix} h_1(\boldsymbol{\chi}_k^{(i)}) & \cdots & h_4(\boldsymbol{\chi}_k^{(i)}) \end{bmatrix}^T, \tag{26}$$

giving the predicted range vector

$$\widehat{\mathbf{z}}_k^- = \sum_{i=0}^{2L} W_m^{(i)} \mathbf{z}_k^{(i)}. \tag{27}$$

The sigma-point measurement covariance and state–measurement cross-covariance are

$$\mathbf{S}_k^{sp} = \sum_{i=0}^{2L} W_c^{(i)} (\mathbf{z}_k^{(i)} - \widehat{\mathbf{z}}_k^-)(\mathbf{z}_k^{(i)} - \widehat{\mathbf{z}}_k^-)^T, \tag{28}$$

$$\mathbf{P}_{xz,k} = \sum_{i=0}^{2L} W_c^{(i)} (\boldsymbol{\chi}_k^{(i)} - \widehat{\mathbf{x}}_k^*)(\mathbf{z}_k^{(i)} - \widehat{\mathbf{z}}_k^-)^T. \tag{29}$$

For the corrected range vector $\mathbf{d}_k = [d_{1,k}, \ldots, d_{4,k}]^T$, the innovation and componentwise correntropy weights are

$$\mathbf{e}_k = \mathbf{d}_k - \widehat{\mathbf{z}}_k^-, \qquad g_{j,k} = G_\sigma(e_{j,k}). \tag{30}$$

With the diagonal nominal covariance $\mathbf{R} = \mathrm{diag}(R_{11}, \ldots, R_{44})$, MCC produces the effective covariance

$$\widetilde{\mathbf{R}}_k = \mathrm{diag}\left(\frac{R_{11}}{\max(g_{1,k}, \epsilon)}, \ldots, \frac{R_{44}}{\max(g_{4,k}, \epsilon)}\right), \tag{31}$$

where $\epsilon > 0$ prevents numerical division by zero. Abnormal distance data therefore receives a larger effective variance and a smaller contribution to the correction. The robust innovation covariance, gain, state, and covariance updates are

$$\mathbf{S}_k^{MCC} = \mathbf{S}_k^{sp} + \widetilde{\mathbf{R}}_k, \qquad \mathbf{K}_k = \mathbf{P}_{xz,k}(\mathbf{S}_k^{MCC})^{-1}, \tag{32}$$

$$\widehat{\mathbf{x}}_k = \widehat{\mathbf{x}}_k^* + \mathbf{K}_k\mathbf{e}_k, \tag{33}$$

The covariance is then updated directly from the sigma-point-based innovation covariance, without linearizing the range measurement model,

$$\mathbf{P}_k = \mathbf{P}_k^* - \mathbf{K}_k\mathbf{S}_k^{MCC}\mathbf{K}_k^T. \tag{34}$$

The process-noise scale is updated only from measurements whose correntropy weights exceed the inlier threshold.

Let $\mathbf{e}_{I,k}$, $\mathbf{H}_{I,k}$, and $\mathbf{R}_{I,k}$ denote the estimation residuals, Jacobian matrix, and nominal measurement covariance restricted to the inlier set. Using the process-propagated covariance $\mathbf{P}_k^p$, the covariance-matching update is

$$a_k = \mathbf{e}_{I,k}^T \mathbf{e}_{I,k} - \mathrm{tr}(\mathbf{H}_{I,k}\mathbf{P}_k^p\mathbf{H}_{I,k}^T) - \mathrm{tr}(\mathbf{R}_{I,k}), \quad (35)$$

$$b_k = \mathrm{tr}(\mathbf{H}_{I,k}\mathbf{Q}\mathbf{H}_{I,k}^T),\ \lambda_k = \mathrm{clip}\left(\frac{a_k}{b_k}, \lambda_{min}, \lambda_{max}\right). \quad (36)$$

Clipping bounds the prediction uncertainty and prevents unstable covariance growth [10].

Algorithm 1 Sensor Fusion Maximum Correntropy Criterion Unscented Kalman Filter

Input: Corrected UWB ranges $\mathbf{d}_k$; encoder speed $v_{enc,k}$; IMU yaw rate $\omega_{z,k}$; anchors; Δt; $\mathbf{Q}$; $\mathbf{R}$; $\mathbf{R}_s$; σ; unscented parameters; adaptation bounds.

Output: Position estimate $[\hat{x}_k, \hat{y}_k]^T$.

if the filter is not initialized **then**
  Average five valid LSE positions.
  Set $\widehat{\mathbf{x}} = [\bar{x}_{LS}, \bar{y}_{LS}, v_{enc}, \omega_z]^T$, initialize $\mathbf{P}$ and θ, and set $\lambda = 1$ {Start with standard SF-UKF}.
end if
Predict $\widehat{\mathbf{x}}^-$, $\mathbf{P}^p$, and $\mathbf{P}^-$ using (13)–(15); propagate θ using (16).
Form $\mathbf{z}_s = [v_{enc}, \omega_z]^T$.
Apply the Gaussian speed/yaw-rate update to obtain $\widehat{\mathbf{x}}^*$ and $\mathbf{P}^*$.
Generate sigma points from $\widehat{\mathbf{x}}^*, \mathbf{P}^*$ and transform them through the four range functions.
Compute $\widehat{\mathbf{z}}^-$, $\mathbf{S}^{sp}$, and $\mathbf{P}_{xz}$.
Compute the innovation $\mathbf{e}$ and correntropy weights g_j.
Form $\widetilde{\mathbf{R}}$ and calculate the robust gain $\mathbf{K}$.
Update $\widehat{\mathbf{x}}$ using (33).
Update $\mathbf{P}$ directly from $\mathbf{K}_k$ and $\mathbf{S}_k^{MCC}$ using (34)
Select the inlier ranges from their correntropy weights.
if at least two inlier ranges are available **then**
  Update and clip λ using (36).
end if
Return $[\hat{x}, \hat{y}]^T$.

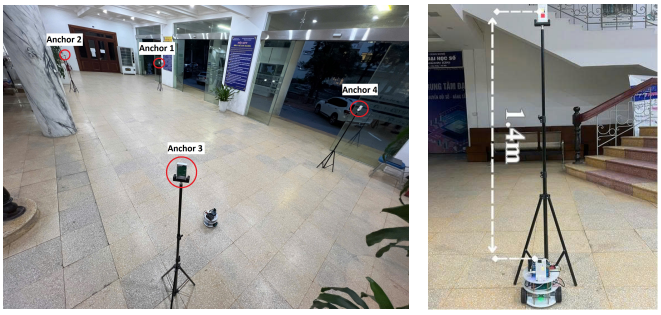
IV. SIMULATION AND EXPERIMENTAL EVALUATION

A. Experimental Setup and Evaluation Protocol

The experiment was conducted at Lecture Hall 2, VNU University of Engineering and Technology. Four DWM1001 anchors covered an $8.8\ \mathrm{m} \times 4.0\ \mathrm{m}$ rectangular area, with one mobile tag acquiring Single-Sided Two-Way Ranging measurements Fig. 2(a). The anchors were mounted 1400 mm above the tag Fig. 2(b), with ground-plane coordinates $\mathbf{A}_1 = (4000, 8800)$, $\mathbf{A}_2 = (0, 8800)$, $\mathbf{A}_3 = (0, 0)$, and $\mathbf{A}_4 = (4000, 0)$ mm.

The AMR followed the rectangular path in Fig. 3 through waypoints $\mathbf{A} = (800, 400)$, $\mathbf{B} = (3000, 400)$, $\mathbf{C} = (3000, 8000)$, and $\mathbf{D} = (800, 8000)$ mm before returning to $\mathbf{A}$. At a mean speed of $0.2\ \mathrm{m/s}$, it recorded synchronized UWB ranges, wheel-encoder speed, and BNO055 yaw rate.

Five methods were compared: LSE and MCC-UKF use only UWB measurements, with the latter incorporating correntropy for robustness; SF-KF and SF-UKF additionally fuse encoder and IMU data; and the proposed SF-MCC-UKF



(a) (b)

Fig. 2. Experimental setup: (a) indoor LOS environment; (b) anchor installation.

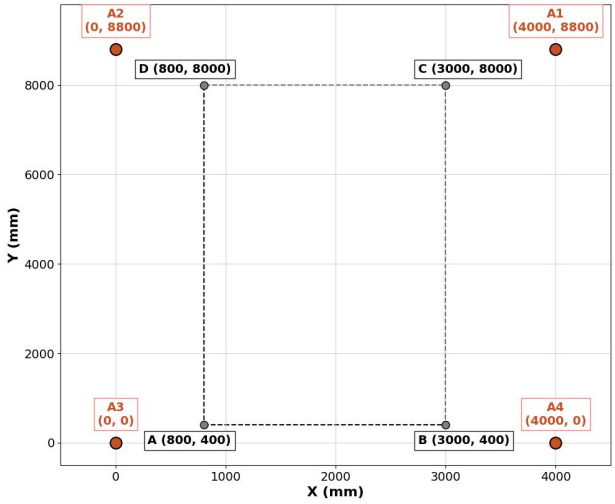


Fig. 3. Rectangular reference trajectory and UWB anchor configuration.

(Algorithm 1) combines this sensor fusion with correntropy-based UWB weighting.

All methods used the same recording and height correction, with recursive filters initialized from the same first five valid LSE samples. Error statistics were computed over the common valid time interval, without method-specific sample removal after inference.

B. Simulation

To validate the proposed algorithm under controlled conditions with significant outliers, a simulation was conducted prior to the physical experiment. The simulated environment modeled a UWB network with aggressive non-Gaussian noise and severe LOS degradation, while only sensor noise was added to the IMU and odometry data to isolate the impact of heavy UWB spikes.

1) Overall Performance and Reliability: As shown in Fig. 4, the severe impact of raw UWB outliers is evident. The baseline LSE method yields a highly unreliable trajectory with an RMSE of 355.9 mm. While MCC-UKF improves robustness against these outliers, with RMSE of 121.1 mm, it lacks the kinematic constraints provided by internal sensors. Conversely, traditional sensor fusion methods fail to fully reject the aggressive noise. Integrating IMU and odometry via SF-KF yields an RMSE of 151.3 mm. Utilizing the

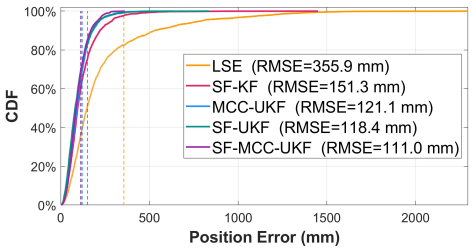


Fig. 4. Simulation: Cumulative Distribution Function (CDF) of position error.

unscented transform, SF-UKF handles system non-linearities better and reduces the RMSE to 118.4 mm; however, its standard formulation remains vulnerable to non-Gaussian spikes. The proposed SF-MCC-UKF successfully bridges this gap, achieving the best overall RMSE of 111.0 mm. Specifically, it improves the positioning accuracy by 68.8%, 26.6%, 8.3%, and 6.3% compared to the LSE, SF-KF, MCC-UKF, and SF-UKF baselines, respectively. Its rapid convergence to 100% on the CDF curve further demonstrates exceptional reliability under aggressive noise.

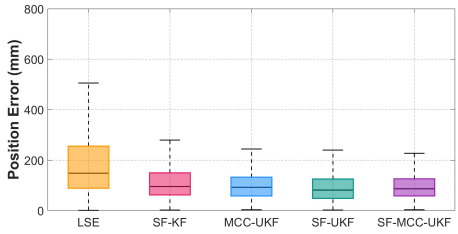


Fig. 5. Simulation: Position error distribution for different localization methods.

TABLE I
SIMULATION SUMMARY STATISTICS OF POSITION ERROR

Algorithm	**RMSE** (mm)	**P50** (mm)	**P95** (mm)	**Max** (mm)	**Std** (mm)
LSE	355.9	148.2	797.1	2295.9	266.0
SF-KF	151.3	95.5	289.5	1450.9	93.7
MCC-UKF	121.1	92.3	218.6	632.2	64.1
SF-UKF	118.4	**81.4**	217.7	835.2	68.6
SF-MCC-UKF	**111.0**	86.4	**208.8**	**360.9**	**54.1**

2) Position Error Distribution: Detailing the data dispersion, the simulation results illustrated in Fig. 5 and Table I confirm that LSE suffers from a massive maximum error of 2295.9 mm. Even with standard sensor fusion, the SF-KF maximum error remains significantly high at 1450.9 mm. While SF-UKF reduces this maximum error to 835.2 mm by better approximating nonlinear transformations, it still suffers heavily from outlier-induced spikes. In contrast, the MCC-UKF demonstrates the power of the correntropy criterion by sharply bounding the maximum error to 632.2 mm and the standard deviation to 64.1 mm, despite the absence of internal sensor fusion. By integrating this robust outlier rejection with vehicle kinematics, the proposed SF-MCC-UKF further minimizes the standard deviation to 54.1 mm and tightly caps the maximum error at an exceptional 360.9 mm. This confirms that the integration of correntropy into the fusion framework is essential for providing superior stability under high-variance noise conditions.

C. Results Analysis

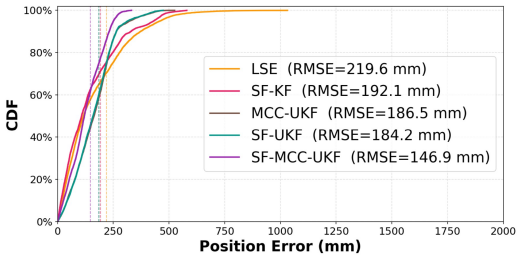


Fig. 6. CDF of position error.

1) Overall Performance and Reliability: As depicted in the CDF chart (Fig. 6), the LSE method exhibits the lowest reliability and highest overall error, with RMSE of 219.6 mm, as directly approximating the position using raw UWB distance data is heavily degraded by measurement spikes. Integrating IMU and odometer data through SF-KF reduces the RMSE to 192.1 mm. The SF-UKF method provides a further improvement, achieving an RMSE of 184.2 mm. This marginal gain indicates that while UKF handles real-world non-linear kinematics better than KF, the primary challenge remains the non-Gaussian measurement noise. Conversely, the MCC-UKF method mitigates these spikes but lacks kinematic constraints, yielding an RMSE of 186.5 mm.

The proposed SF-MCC-UKF algorithm demonstrates absolute superiority by optimally combining outlier rejection with internal vehicle kinematics. It achieves the lowest overall RMSE of 146.9 mm. This translates to a significant accuracy improvement of 33.1%, 23.5%, 21.2%, and 20.2% over the traditional LSE, SF-KF, MCC-UKF, and SF-UKF baselines, respectively. Furthermore, the purple curve of the proposed method converges to 100% most rapidly. Tracking the 95% cumulative probability mark, the LSE and SF-KF curves indicate errors exceeding 430 mm, whereas the SF-MCC-UKF curve successfully bounds the error to approximately 250 mm. By automatically scaling the measurement noise covariance matrix, the correntropy mechanism ensures that severe physical UWB outliers do not corrupt the dynamic state estimation.

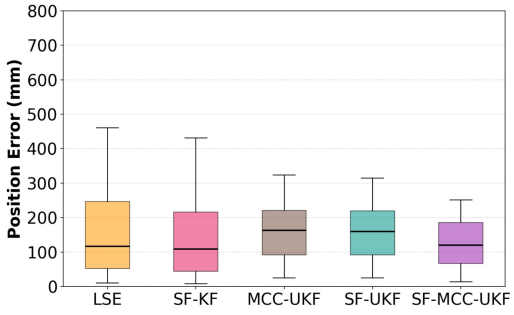


Fig. 7. Position error distribution for different localization methods.

2) Position Error Distribution: As shown in Fig. 7 and Table II, the proposed SF-MCC-UKF yields the most stable and concentrated error distribution. The LSE baseline suffers from a severe maximum error of 1031.3 mm. While the MCC-UKF effectively suppresses extreme error spikes, lowering the maximum error to 525.4 mm, and SF-KF, SF-UKF utilize vehicle kinematics to tighten the standard deviation, they all remain vulnerable to outliers. By integrating correntropy updates with sensor fusion, SF-MCC-UKF substantially outperforms these baselines, restricting the maximum error to only 332.0 mm and minimizing the standard deviation to 74.4 mm.

TABLE II
EXPERIMENTAL SUMMARY STATISTICS OF POSITION ERROR

Algorithm	**RMSE** (mm)	**P50** (mm)	**P95** (mm)	**Max** (mm)	**Std** (mm)
LSE	219.6	116.0	461.0	1031.3	145.8
SF-KF	192.1	**108.0**	431.0	580.0	127.1
MCC-UKF	186.5	163.0	323.9	525.4	91.4
SF-UKF	184.2	159.0	314.5	474.1	90.0
SF-MCC-UKF	**146.9**	120.0	**251.0**	**332.0**	**74.4**

Furthermore, although the median error of SF-KF is 108.0 mm, which is slightly lower than the 120.0 mm achieved by SF-MCC-UKF, the proposed algorithm exhibits significantly higher overall stability. This is explicitly evidenced by its sharp reductions in the 95th percentile and maximum errors, highlighting the indispensable role of the MCC framework in suppressing variance caused by large, sudden environmental noise.

3) Simulation vs. Experimental Comparison: The physical experiment introduces additional effects such as wheel slippage, vibration, IMU drift, and unmodeled UWB disturbances, leading to different error distributions from the simulation. Nevertheless, SF-MCC-UKF achieves the lowest RMSE and effectively limits extreme errors in both evaluations, indicating that correntropy-based UWB weighting remains beneficial when combined with kinematic sensor fusion.

V. CONCLUSION AND FUTURE WORK

This paper has proposed a robust indoor localization approach for autonomous mobile robots using SF-MCC-UKF. The proposed algorithm integrates nonlinear sigma-point fusion of UWB, IMU, and odometry data with a correntropy-based measurement update to effectively mitigate the impact of non-Gaussian noise and severe measurement outliers.

Under the evaluated conditions, SF-MCC-UKF achieved an RMSE of 111.0 mm in simulation and 146.9 mm in the physical experiment. The experimental RMSE was 20.2–33.1% lower than those of LSE, SF-KF, MCC-UKF, and SF-UKF, while the P95 and maximum errors were limited to 251.0 mm and 332.0 mm, respectively.

Future work will evaluate the method under more challenging NLOS conditions and obstacle configurations and investigate data-driven UWB error compensation and collaborative multi-robot localization.

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